

Sumit Sharma

AI/ML Engineer | B.Tech – Computer Science & Engineering (Data Science)
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PROFESSIONAL SUMMARY

AI/ML Engineer and B.Tech CSE (Data Science) student who builds production Generative AI systems end to end: LoRA fine-tuning of Llama 3.2 with Hugging Face PEFT, Retrieval-Augmented Generation (RAG) over pgvector and FAISS, Whisper speech-to-text pipelines, and LLM agents with intent routing, tool use and token streaming. Strong in Python, PyTorch, Transformers, LangChain, prompt engineering, FastAPI and cloud deployment (Docker, Google Cloud Run, CI/CD); two published research papers.

EXPERIENCE

Web Development Intern (Full-Stack & AI Assistant)

Jan 2025 – Present

ITM Gwalior

Gwalior, India

- Built and shipped the official ITM Gwalior website, enterprise CMS and RAG-based AI assistant (React, FastAPI, PostgreSQL, Redis): 340+ REST endpoints, 70 SQLAlchemy models, 65 public pages and 17 admin modules.
- Implemented RBAC with 60 granular scopes and 22 role presets, JWT access/refresh tokens with rotation, Argon2 hashing, brute-force lockout, audit logging and Redis-backed rate limiting, covered by a pytest suite.
- Owned deployment: Dockerized API on Google Cloud Run via Cloud Build CI/CD, Supabase PostgreSQL, Cloudflare R2 storage, Redis read-through caching, and a prerendered React frontend on Vercel.

Data Analytics Intern

May 2025 – Jun 2025

Navodita Infotech

Remote

- Performed data cleaning, preprocessing and exploratory data analysis (EDA) on client datasets in Python (Pandas, NumPy) and delivered visualizations and reports that informed data-driven business decisions.

PROJECTS

IthaasAI – Fine-Tuned Indian History Chatbot | Llama 3.2, LoRA/PEFT, Hugging Face, PyTorch, FAISS, FastAPI

- Fine-tuned **Llama-3.2-3B-Instruct** with LoRA (rank 16, alpha 32, dropout 0.05, all 7 attention/MLP projection modules) on curated Indian-history data; published the adapter to Hugging Face Hub and built a loader that resolves base model plus PEFT adapter with CUDA/MPS/CPU selection and fp16 inference.
- Paired the model with RAG: 500/50 recursive chunking, all-MiniLM-L6-v2 embeddings and FAISS top-3 retrieval with source citations and relevance scores; token-by-token SSE streaming, 10-turn session memory, Docker Compose deployment, Colab/Kaggle T4 GPU serving.

ITM Gwalior AI Assistant – Production RAG Agent | LangChain, Sentence Transformers, pgvector, Llama 3.1 70B

- Built the ingestion pipeline: 11 scrapers (24 public JSON APIs, BFS website crawler, PDF ingestor), paragraph-aware 512-word chunks with 64-word overlap, and 384-dim all-MiniLM-L6-v2 embeddings in PostgreSQL/pgvector with an HNSW cosine index; migrated from ChromaDB to fix Cloud Run cold-start data loss.
- Designed a zero-LLM-cost intent router (18 categories, 118 keywords) dispatching queries to a SQL tool, category-filtered semantic search or direct chat, with cascading retrieval fallbacks, anti-hallucination guardrails, exponential-backoff retries and SSE token streaming with page-link suggestions.

VNotes – AI YouTube Notes Generator | FastAPI, PostgreSQL, Groq Whisper, GPT-OSS-120B, Docker, Cloud Run

- Built an end-to-end transcript-to-notes pipeline: YouTube captions with auto-translation to English, fallback to audio download and Groq **whisper-large-v3** (domain-biased vocabulary prompt, ffmpeg compression to the 25 MB limit), then prompt-engineered LLM generation of LaTeX-aware Markdown notes, chapters, summaries and action items.
- Engineered for reliability: background jobs with a 7-state status machine, backoff retries, dual LLM providers (Groq, NVIDIA Llama-3.1-70B), tiered daily usage quotas (HTTP 429), Google OAuth to JWT auth; deployed on Google Cloud Run.

TECHNICAL SKILLS

Languages: Python, SQL, C++, C, JavaScript/TypeScript

Machine Learning & Deep Learning: PyTorch, Hugging Face Transformers, PEFT/LoRA Fine-Tuning, Sentence Transformers, Embeddings, NLP, Pandas, NumPy, Matplotlib, EDA

Generative AI & LLMs: RAG, LangChain, Prompt Engineering, LLM Agents & Tool Use, Guardrails, SSE Streaming, Whisper, Groq, NVIDIA NIM, Gemini, OpenAI

Databases & Vector Stores: PostgreSQL, pgvector, FAISS, ChromaDB, Redis, MySQL, Supabase

Backend & MLOps: FastAPI, REST APIs, Pydantic, SQLAlchemy, Alembic, JWT/OAuth 2.0, Docker, Google Cloud Run, CI/CD, Git, Linux, pytest, React, Tailwind CSS

EDUCATION

Institute of Technology and Management, Gwalior

2023 – 2027

B.Tech – Computer Science & Engineering (Data Science) | CGPA: 7.50

Gwalior, India

PUBLICATIONS & CERTIFICATIONS

- Research Papers (IJAMRED, 2025):** Neuron Lifecycle in Deep Networks; Data-Driven Sustainability: Cognitive Decision Intelligence to Reduce Medical Waste in Pharmaceutical Logistics
- Research Chapter:** IIP International Publication (2026) | **Certification:** Oracle Fusion AI Agent Studio Foundations Associate (Rel 1)